

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 701 Cybersecurity — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Anderson — Security Engineering, 3rd edition (free PDF)	The primary source, and free in full from the author. The best book on this subject by a wide margin, and it is about systems and incentives rather than about tools. Modules 01, 11, and 12 follow it closely.
Shostack — Threat Modeling: Designing for Security	Module 01's method. The four-question framework used throughout this course comes from here. Library copy; the author's free materials cover the core.
OWASP — the Application Security Verification Standard and the Cheat Sheet Series (free)	The practical reference for Modules 02–06 — specific, current, defensive, and maintained. The ASVS is the checklist to review a system against.
NIST SP 800-63 (digital identity) and SP 800-53 (controls) (free)	Module 02's authentication guidance, free — and notable for having reversed the industry's password advice on the basis of evidence, which is Module 11's point.
Google — Building Secure and Reliable Systems (free PDF)	Free in full, and the reference for Modules 05, 09, and 10 — security from the perspective of people operating large systems rather than auditing them.
Allspaw and the blameless post-mortem literature (free)	Module 10's practice. Short, and it changed how the industry handles incidents — which is a human-factors result rather than a technical one.

Module-by-module map

Module 01 — Thinking About Security

Resource	What it covers for this module
Anderson — Security Engineering, chapters 1–3 (free PDF)	The asymmetry, the principles, and the economics — free in full, and the first chapter is the best framing of this subject in print.
Shostack — Threat Modeling, and the free four-question materials	§2's method , including STRIDE per element and the trust-boundary guidance.
Saltzer & Schroeder — The Protection of Information in Computer Systems (free)	§3's principles, in the 1975 original — least privilege, fail-safe defaults, and open design were all stated here, and the paper reads as current.
OWASP — Threat Modeling Cheat Sheet (free)	§2 as a working procedure , with the common mistakes listed.

Module 02 — Authentication

Resource	What it covers for this module
NIST SP 800-63B — Digital Identity Guidelines, Authentication (free)	§2's current guidance , including the reversal of the composition and rotation rules, with the reasoning.
OWASP — Authentication, Password Storage, and Session Management Cheat Sheets (free)	The working reference for every section of this module , kept current.
The WebAuthn specification and passkeys documentation (free)	§1's origin-bound factor — and the origin-binding mechanism is worth reading, because it is what makes the phishing resistance work.
Bonneau et al. — The Quest to Replace Passwords (free)	A systematic comparison of authentication schemes on usability, deployability, and security — and the framework explains why passwords persisted.

Module 03 — Authorisation

Resource	What it covers for this module
OWASP — Authorization and Access Control Cheat Sheets, and the Top Ten A01 (free)	The working reference for §3 and §4 , and A01 documents why this is the top-ranked class.
Hardy — The Confused Deputy (free)	§2's pattern, named — two pages, from 1988, and it explains five modern vulnerability classes.
Pang et al. — Zanzibar: Google's Consistent, Global Authorization System (free)	§1's ReBAC model at scale , and a good illustration of the chokepoint principle implemented as a service.
Saltzer & Schroeder, and Miller's capability work (free)	Least privilege and the capability alternative to ambient authority , which is §2's structural fix.

Module 04 — Network Security

Resource	What it covers for this module
Anderson — Security Engineering, the networking chapters (free PDF)	§1 and §3 in context , with the history of why the perimeter model was adopted and why it failed.
Georgiev et al. — The Most Dangerous Code in the World (free)	§1's certificate validation failures, surveyed across real software — and the title is not hyperbole. Required reading.
NIST SP 800-207 — Zero Trust Architecture (free)	§3 without the marketing — the architectural components and the sequencing.
Verizon Data Breach Investigations Report (free, annual)	§4's incident data — read the initial-access section each year, and compare it against where your budget goes.

Module 05 — Platform and Isolation

Resource	What it covers for this module
Google — Building Secure and Reliable Systems, the isolation chapters (free PDF)	§1 and §2 from people operating this at scale , free in full.
Provos, Friedl & Honeyman — Preventing Privilege Escalation (free)	§2's design, in the paper that introduced it to OpenSSH — and the design rules are stated explicitly.
Szekeres et al. — SoK: Eternal War in Memory (free)	§3's classes and mitigations, systematised — which attack each mitigation stops and which it does not.
Kocher et al. — Spectre Attacks (free)	§4's callout — and worth reading for the argument that the boundary was defined at the wrong level.

Module 06 — Web Security

Resource	What it covers for this module
OWASP — the Cheat Sheet Series and the Top Ten (free)	The working reference for this entire module — specific, current, and defensive, with a sheet per class.
Zalewski — The Tangled Web	§1's model, explained properly — the best account of how the browser security model came to be what it is. Library copy; the author's Browser Security Handbook is free.
Google — CSP: A Successful Mess Between Hardening and Mitigation (free)	§4's honest assessment of CSP , from people who deployed it at scale — including why strict policies are hard.
MDN — Same-origin policy, CORS, and SameSite cookies (free)	The accurate current reference for §1 and §3 , which matters because the defaults have changed recently.

Module 07 — Secrets and Keys

Resource	What it covers for this module
OWASP — Secrets Management Cheat Sheet (free)	The working reference for §1 and §4 , including the storage hierarchy and the scanning guidance.
Google — Building Secure and Reliable Systems, the key management material (free PDF)	§2's bootstrap problem and §3's rotation , from people who operate it — and the identity-rather-than-secret framing is theirs.
Meli, McNiece & Reaves — How Bad Can It Get? (free)	§1's committed-secret problem, measured — the scale is larger than you would guess.
NIST SP 800-57 — Recommendation for Key Management (free)	§3's rotation and lifetime guidance , and the key-state model is worth borrowing even if the document is long.

Module 08 — The Supply Chain

Resource	What it covers for this module
SLSA — Supply-chain Levels for Software Artifacts (free)	§2 and §3 as a graded framework — the levels are a usable roadmap rather than a checklist, and the build/release separation is explicit.
Ohm et al. — Backstabber's Knife Collection (free)	§1's risks, from real incidents — a systematic study of malicious package attacks and the mechanisms they used.
The Reproducible Builds project (free)	§3's strongest mechanism , with the practical guidance on the sources of non-determinism.
OpenSSF — Scorecard and the Concise Guides (free)	§1's adoption checks, automated — maintenance activity, review practices, and signing, scored per dependency.

Module 09 — Detection

Resource	What it covers for this module
Google — Building Secure and Reliable Systems, the detection chapters (free PDF)	§1 and §4 from operators — including the logging-design-backwards argument.
Axelsson — The Base-Rate Fallacy and Intrusion Detection (free)	§3, with the arithmetic — the paper that should have settled how alerting is designed and did not.
MITRE ATT&CK (free)	A catalogue of adversary techniques with detection guidance per technique — useful for §2's rule design, and best used threat-model-first rather than as a checklist.
Atomic Red Team (free)	§2's detection testing, automated — small, safe tests that trigger specific detections, for use on your own systems.

Module 10 — Incident Response

Resource	What it covers for this module
NIST SP 800-61 — Computer Security Incident Handling Guide (free)	§1's phases, in detail , with the preparation checklists worth borrowing.
Allspaw — Blameless PostMortems and a Just Culture (free)	§4's practice , short, and it changed how the industry handles incidents.
Dekker — The Field Guide to Understanding Human Error	§4's framing, from safety engineering — where the human-error-as-symptom argument originates. Library copy.
Google SRE Book — the incident management chapters (free)	§1's roles and command structure , free in full — and the incident-commander model is worth adopting.

Module 11 — Human Factors

Resource	What it covers for this module
Anderson — Security Engineering, the psychology and usability chapters (free PDF)	§1 and §2 with the evidence — free in full, and the strongest treatment of this material anywhere.
Beautement, Sasse & Wonham — The Compliance Budget (free)	§1's second callout, in the original — and the framing is more useful than any individual finding.
Adams & Sasse — Users Are Not the Enemy (free)	§1's argument, from 1999 , and the title is the thesis. It took the industry twenty years to act on it.
Thaler & Sunstein — Nudge; and the default-effect literature	§3's effect, outside security — where the default-dominance finding was established, and it transfers directly.

Module 12 — Measuring Security

Resource	What it covers for this module
Anderson — Security Engineering, the economics chapter (free PDF)	§4's incentive problems, developed properly — and the externality argument is the most important idea in this module.
Hubbard & Seiersen — How to Measure Anything in Cybersecurity Risk	§3's case for quantification , argued at length — read alongside this module's caveat about absolute figures. Library copy.
Verizon DBIR and the Cyentia reports (free)	The nearest thing to population-level data — which is what individual organisations lack (§1).
Goodhart, and Muller — The Tyranny of Metrics	§1's table as a general phenomenon — the same distortion documented across medicine, education, and policing.

Module 13 — Claiming Security Honestly

Resource	What it covers for this module
Anderson — Security Engineering, the concluding chapters (free PDF)	§3 and §4 in context , and the best available account of why programmes are ordered the way they are rather than the way they should be.
OWASP — Application Security Verification Standard (free)	A levelled checklist to review a system against , with the levels making the proportionality of §4 explicit.
CIS Critical Security Controls (free)	§4's ordering, as a published prioritisation — and the implementation groups are a proportionality framework worth borrowing.
Verizon DBIR, the initial-access section (free, annual)	The evidence behind §4's ordering — reread it annually and check your programme against it.